$$\frac{\Gamma \vdash e_1 : \tau \to \mathsf{bool} \rightarrowtail^{\mathrm{d}} e_3 \qquad \Gamma \vdash e_2 : \tau \rightarrowtail^{\mathrm{d}} e_4}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\mathsf{flat}(e_1), e_2) : \tau \rightarrowtail^{\mathrm{d}} \mathtt{perform}\ \mathscr{C}\ \tau\ \langle k, \langle e_3, e_4 \rangle\rangle} \text{ Compile-Flat}$$

$$\frac{\begin{array}{c}\Gamma \vdash e_1 : \tau_2 \rightarrowtail^{\mathrm{d}} e_2 \\ \Gamma \vdash \lambda f : \tau_1 \to \tau_2 . \lambda x : \tau_1 . \mathsf{mon}_j^{k,l}\left(\kappa_2, f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail^{\mathrm{d}} e_3\end{array}}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to \kappa_2, e_1) : \tau_1 \to \tau_2 \rightarrowtail^{\mathrm{d}} e_3\ e_2} \text{ Compile-Func}$$

$$\frac{\begin{array}{c}f, x \notin \mathrm{dom}(\Gamma) \cup \{y\} \cup \mathrm{fv}(\kappa_1, \kappa_2) \qquad \Gamma \vdash e_1 : \tau_1 \to \tau_2 \rightarrowtail^{\mathrm{d}} e_2 \\ e_w = \lambda f : \tau_1 \to \tau_2 . \lambda x : \tau_1 . \mathsf{mon}_j^{k,l}\left(\kappa_2\left[y := \mathsf{mon}_j^{l,j}(\kappa_1, x)\right], f\ \mathsf{mon}_j^{l,k}(\kappa_1, x)\right) \\ \Gamma \vdash e_w : (\tau_1 \to \tau_2) \to (\tau_1 \to \tau_2) \rightarrowtail^{\mathrm{d}} e_3\end{array}}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\kappa_1 \to_i \lambda y . \kappa_2, e_1) : \tau_1 \to \tau_2 \rightarrowtail^{\mathrm{d}} e_3\ e_2} \text{ Compile-Dep}$$

$$\frac{\begin{array}{c}x \notin \mathrm{dom}(\Gamma) \qquad \Gamma, x : \langle \tau_1, \tau_2 \rangle \vdash \mathsf{mon}_j^{k,l}(\kappa_1, \mathsf{inj.l}(x)) : \tau_1 \rightarrowtail^{\mathrm{d}} e_1 \\ \Gamma, x : \langle \tau_1, \tau_2 \rangle \vdash \mathsf{mon}_j^{k,l}(\kappa_2, \mathsf{inj.r}(x)) : \tau_2 \rightarrowtail^{\mathrm{d}} e_2 \qquad \Gamma \vdash e : \langle \tau_1, \tau_2 \rangle \rightarrowtail^{\mathrm{d}} e_3\end{array}}{\Gamma \vdash \mathsf{mon}_j^{k,l}(\langle \kappa_1, \kappa_2 \rangle, e) : \langle \tau_1, \tau_2 \rangle \rightarrowtail^{\mathrm{d}} (\lambda x . \langle e_1, e_2 \rangle)\ e_3} \text{ Compile-Tuple}$$

$$\mathtt{h}_{\mathscr{C}} : \mathtt{lab} \quad = \quad \{\mathscr{C} \to \forall \alpha . \langle \mathtt{blab}, \langle \alpha \to \mathsf{bool}, \alpha \rangle\rangle \to \alpha\}$$

$$\mathsf{wrap} \quad = \quad \mu w : \forall \alpha . \left(\left(() \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\right) \to \alpha\right) . \Lambda \alpha .\ \lambda f : \left(() \to^{\langle \mathtt{h}_{\mathscr{C}} \rangle} \alpha\right) .$$

$$\mathtt{handler}_d(\{\mathscr{C} \to$$

$$\lambda \langle l, \langle e, v \rangle\rangle . \lambda k .\ \text{if } w[\mathsf{bool}]\ (\lambda _ . e\ v) \text{ then } k\ v \text{ else } \mathtt{error}_l\}) \ f$$

where $\lambda \langle l, \langle e, v \rangle\rangle$ deconstructs the argument as tuple pattern

Figure 1: Compiler Rules